THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — MATHS · CLASS 9 · SET 05

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. The point which lies on both the x-axis and the y-axis is:

(a) (1, 0) (b) (0, 1) (c) (1, 1) (d) (0, 0)

Ans: (d) Only the origin has both coordinates zero.

2. The ordinate of the point (−2, 9) is:

(a) −2 (b) 9 (c) −9 (d) 7

Ans: (b) Ordinate = y-coordinate = 9.

3. The slope of the line $y = 2x + 3$ is:

(a) 2 (b) 3 (c) 1/2 (d) −2

Ans: (a) In the slope-intercept form $y = mx + c$ the slope is $m = 2$.

4. Which of the following is a rational number?

(a) $\sqrt{2} \times \sqrt{3}$ (b) $\sqrt{20} \times \sqrt{5}$ (c) $\pi + 1$ (d) $\sqrt{7} + \sqrt{3}$

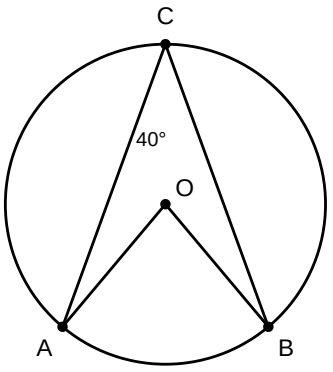
Ans: (b) $\sqrt{20} \times \sqrt{5} = \sqrt{100} = 10$ is rational; $\sqrt{6}$, $\sqrt{7} + \sqrt{3}$ and $\pi + 1$ are irrational.

5. If $x + 1/x = 5$, then $x^2 + 1/x^2$ is equal to:

(a) 21 (b) 27 (c) 25 (d) 23

Ans: (d) $(x + 1/x)^2 = x^2 + 2 + 1/x^2 \Rightarrow x^2 + 1/x^2 = 25 - 2 = 23$.

6. In the figure, O is the centre of the circle and $\angle ACB = 40°$. Then $\angle AOB$ is equal to:



(a) 80° (b) 20° (c) 40° (d) 160°

Ans: (a) Angle at the centre is twice the angle at the circumference: $2 \times 40° = 80°$.

7. A chord is 12 cm from the centre of a circle and its length is 10 cm. The radius of the circle is:

(a) 11 cm (b) $\sqrt{119}$ cm (c) 17 cm (d) 13 cm

Ans: (d) Half chord = 5 cm; $r^2 = 12^2 + 5^2 = 169 \Rightarrow r = 13$ cm.

8. The area of an equilateral triangle of perimeter 36 cm, by Heron's formula, is:

(a) 144 cm² (b) $18\sqrt{3}$ cm² (c) $36\sqrt{3}$ cm² (d) $72\sqrt{3}$ cm²

Ans: (c) Side = 12 cm, s = 18; area = $\sqrt{(18 \times 6 \times 6 \times 6)} = 36\sqrt{3}$ cm².

9. A day of the week is chosen at random. The probability that its name begins with the letter 'T' is:

(a) $\frac{3}{7}$ (b) $\frac{1}{7}$ (c) $\frac{1}{2}$ (d) $\frac{2}{7}$

Ans: (d) Tuesday and Thursday ⇒ 2/7.

10. A sequence is defined by $t_1 = 1$ and $t_n = t_{n-1} + n$ for $n \geq 2$. Then t_5 is:

(a) 15 (b) 12 (c) 10 (d) 21

Ans: (a) 1, 3, 6, 10, 15.

11. The converse of 'If $x = y$, then $x^3 = y^3$' is:

(a) If $x = y$, then $x^2 = y^2$. (b) If $x^3 = y^3$, then $x^2 = y^2$.
(c) If $x^3 = y^3$, then $x = y$. (d) If $x \neq y$, then $x^3 \neq y^3$.

Ans: (c) Interchange the two parts. (For real numbers this converse is also true.)

12. Euler disproved Fermat's claim that all numbers of the form $2^{2^n} + 1$ are prime by showing that the number obtained for n = 5 is composite. That number is:

(a) $2^{32} + 1$ (b) $2^4 + 1$ (c) $2^8 + 1$ (d) $2^{16} + 1$

Ans: (a) For n = 5, 2^(2⁵) + 1 = $2^{32} + 1$ is composite. The numbers for n = 0 to 4 (3, 5, 17, 257, 65537) are prime.

13. Section A has 8 students with an average score of 40 and Section B has 12 students with an average score of 50. The combined average is:

(a) 45 (b) 47 (c) 44 (d) 46

Ans: (d) $(8 \times 40 + 12 \times 50)/20 = 920/20 = 46$.

14. A cylinder and a cone have the same base radius and the same height. The ratio of the volume of the cylinder to that of the cone is:

(a) 2 : 1 (b) 3 : 1 (c) 1 : 3 (d) 3 : 2

Ans: (b) $\pi r^2 h : \frac{1}{3}\pi r^2 h = 3 : 1$.

15. The final gcd algorithm checks every k from 2 to min(m, n). For m = 20 and n = 50, the number of values of k checked is:

(a) 19 (b) 48 (c) 20 (d) 49

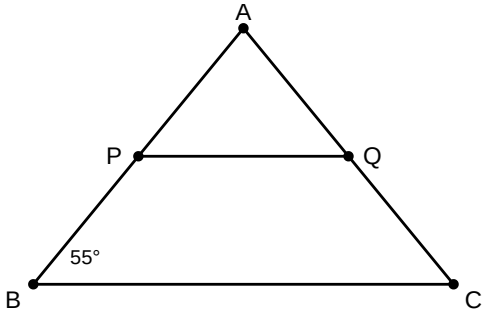
Ans: (a) k = 2, 3, …, 20 ⇒ 19 values.

16. A quadrilateral whose diagonals bisect each other is always a:

(a) rectangle (b) trapezium (c) parallelogram (d) rhombus

Ans: (c) Diagonals bisecting each other is a test for a parallelogram.

17. In the figure, P and Q are the midpoints of AB and AC. If ∠ABC = 55°, then ∠APQ is:



(a) 55° (b) 35° (c) 70° (d) 125°

> **Ans: (a)** PQ ∥ BC (Midpoint Theorem), so corresponding angles are equal: ∠APQ = ∠ABC = 55°.

18. The sum of the first 10 terms of the AP 3, 7, 11, … is:

(a) 220 (b) 200 (c) 230 (d) 210

> **Ans: (d)** $S_{10} = 10/2 \times [2(3) + 9(4)] = 5 \times 42 = 210$.

19. **Assertion (A):** $(a + b)^2 = a^2 + b^2$ for all real numbers a and b.
Reason (R): $(a + b)^2 = a^2 + 2ab + b^2$.

> **Ans: (d)** A is false (e.g. a = b = 1 gives 4 ≠ 2); R is the true identity.

20. **Assertion (A):** The number 36 has an odd number of factors.
Reason (R): 36 is a perfect square, and a perfect square has an odd number of factors.

> **Ans: (a)** 36 has the 9 factors 1, 2, 3, 4, 6, 9, 12, 18, 36. The factor pairs have the single repeated pair (6, 6) – so R explains A.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

21. A mobile data plan costs ₹99 and includes 1 GB. Every additional GB costs ₹15. Find the cost of using 6 GB.

> Extra data = 6 − 1 = 5 GB (½)
> Extra cost = 5 × 15 = ₹75 (1)
> Cost = 99 + 75 = **₹174** (½)

22. Without actual division, show that $\frac{21}{400}$ has a terminating decimal expansion and state the number of decimal places.

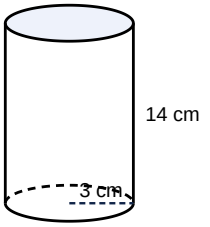
> 400 = $2^4 \times 5^2$ has only 2 and 5 as prime factors ⇒ terminates (1). Number of decimal places = **4** (0.0525) (1)

O R

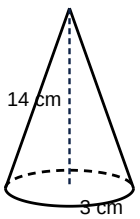
Find two rational numbers between $\frac{-1}{2}$ and $\frac{1}{4}$.

> −1/2 = −2/4, so any two numbers in between, e.g. **−1/4 and 0** (1 each)

23. A solid metal cylinder of radius 3 cm and height 14 cm is melted and recast into cones, each of radius 3 cm and height 14 cm. How many cones are made? Give a reason.



Solid cylinder



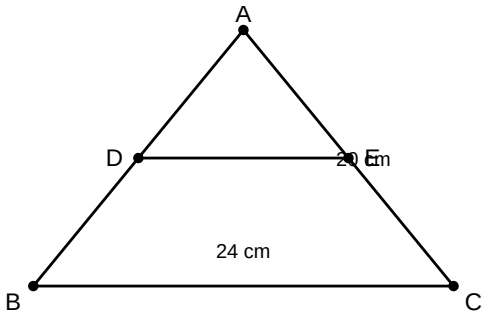
Cone (one of several)

> Volume of cylinder = $\pi r^2 h$; volume of each cone = $\frac{1}{3}\pi r^2 h$ (1)
> Number of cones = $\pi r^2 h \div \frac{1}{3}\pi r^2 h$ = **3** (melting keeps the volume the same) (1)

24. Using the digit-by-digit addition algorithm, add 95 and 67. What would the answer be if the last step (writing the final carry) were omitted?

> 5 + 7 = 12 → write 2, carry 1; 9 + 6 + 1 = 16 → write 6, carry 1; final carry written: **162** (1)
> Without the last step the answer would be **62** (1)

25. In the figure, D is the midpoint of AB and DE ∥ BC. If BC = 24 cm and AC = 20 cm, find DE and AE, giving reasons.



By the converse of the Midpoint Theorem E is the midpoint of AC ⇒ **AE = 10 cm** (1)
By the Midpoint Theorem **DE = ½BC = 12 cm** (1)

SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Express 1.2424… (24 repeating) in the form p/q in lowest terms.

x = 1.2424…; 100x = 124.2424… (1)
99x = 123 (1)
x = 123/99 = **41/33** (1)

27. In a survey of 150 people, their favourite drink was recorded:

Drink	Tea	Coffee	Juice	None
People	60	45	30	15

A person is chosen at random. Find the probability that the person (i) prefers tea, (ii) prefers coffee or juice, (iii) has no favourite.

Total = 150. (i) 60/150 = **2/5** (1) (ii) 75/150 = **1/2** (1) (iii) 15/150 = **1/10** (1)

O R

A letter is chosen at random from the word ‘PROBABILITY’. Find the probability that it is (i) a vowel, (ii) the letter B, (iii) not the letter I.

11 letters. (i) Vowels O, A, I, I = 4 ⇒ **4/11** (1) (ii) B occurs twice ⇒ **2/11** (1) (iii) I occurs twice ⇒ 9/11 (1)

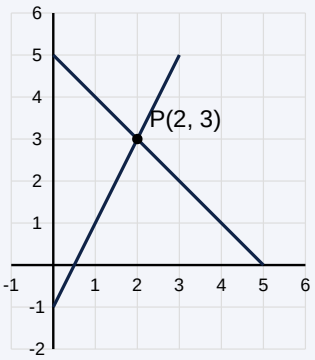
28. Draw the graphs of y = 2x − 1 and x + y = 5 on the same grid. Where do the lines meet? Write the solution of the pair.

Table for y = 2x − 1:

x	0	2
y	-1	3

Table for x + y = 5:

x	0	5
y	5	0



(1) Draw both lines (1) The lines meet at **(2, 3)**, so x = 2, y = 3 (1)

29. Give a counterexample to each claim: (i) ‘$n^2 - n + 41$ is prime for every positive integer n’; (ii) ‘If n is even, then $2^n + 1$ is prime’.

(i) n = 41: $41^2 - 41 + 41 = 41^2 = 1681$ is not prime (1½)
(ii) n = 6: $2^6 + 1 = 65 = 5 \times 13$ is not prime (1½)

30. Using Heron’s formula, find the area of a triangle whose sides are 12 cm, 35 cm and 37 cm.

s = (12 + 35 + 37)/2 = 42 cm (1)
Area = √[42 × 30 × 7 × 5] (1) = √44100 = **210 cm²** (1)

OR

A chord AB of length 18 cm is 12 cm from the centre of a circle. Find the radius, and the length of another chord CD of the same circle that is 9 cm from the centre.

$r^2 = 12^2 + 9^2 = 225$ ⇒ **r = 15 cm** (1)
Half of CD = √(225 − 81) = 12 (1) ⇒ **CD = 24 cm** (1)

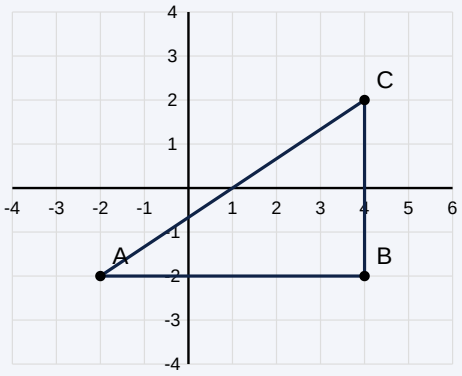
31. The diagonals of a parallelogram ABCD are equal. Show that ABCD is a rectangle.

In △ABC and △DCB: AB = DC (opposite sides), BC = CB (common), AC = DB (given) ⇒ △ABC ≅ △DCB (SSS) ⇒ ∠ABC = ∠DCB (1½)
AB ∥ DC, so ∠ABC + ∠DCB = 180° ⇒ each angle = 90° (1)
A parallelogram with a right angle is a rectangle. (½)

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. A triangular signboard has its corners at A(−2, −2), B(4, −2) and C(4, 2) on a grid.
(a) Plot the points and join them. (2 marks)
(b) Find the lengths AB, BC and AC. (1½ marks)
(c) Use the converse of the Baudhāyana–Pythagoras theorem to show that the triangle is right-angled and find its area. (1½ marks)



(a) Plotting 3 points (1½), joining (½): (b) AB = 6, BC = 4, AC = $\sqrt{(6^2 + 4^2)} = \sqrt{52}$ (1½)
(c) $AB^2 + BC^2 = 36 + 16 = 52 = AC^2$ ⇒ the triangle is right-angled at B (converse of the theorem) (1). Area = ½ × 6 × 4 = **12 square units** (½)

33. Find the sum of all natural numbers between 50 and 250 which leave remainder 3 when divided by 5. (How many such numbers are there? 2½ + 2½)

Numbers: 53, 58, …, 248 — an AP with a = 53, d = 5, l = 248 (1)
n = (248 − 53)/5 + 1 = **40** (1½)
Sum = 40/2 × (53 + 248) = 20 × 301 = **6020** (2½)

O R

The marks of 200 students in an examination were recorded:

Marks	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100
Students	16	44	80	40	20

Find the probability that a student chosen at random scored (i) 60 marks or more (1), (ii) less than 40 marks (1), (iii) 40 or more but less than 80 marks (2), and (iv) comment on the sum of the probabilities of the five classes (1).

Total = 200. (i) (40 + 20)/200 = **3/10** (1) (ii) (16 + 44)/200 = **3/10** (1) (iii) (80 + 40)/200 = **3/5** (2) (iv) The five probabilities 2/25, 11/50, 2/5, 1/5, 1/10 add up to **1** (1)

34. Prove that if the opposite sides of a quadrilateral are equal, then it is a parallelogram.

Let ABCD be a quadrilateral with AB = CD and AD = BC. Join AC. (1)
In △ACD and △CAB: AC = CA (common), CD = AB, AD = CB ⇒ △ACD ≅ △CAB (SSS) (1½)
So ∠DAC = ∠BCA ⇒ AD ∥ BC (alternate angles) (1)
and ∠DCA = ∠BAC ⇒ DC ∥ AB (1)
Both pairs of opposite sides are parallel ⇒ **ABCD is a parallelogram**. (½)

O R

Batch A of brass weighs 100 kg and is 80% copper; Batch B weighs 200 kg and is 50% copper. Batch C is 30% copper. The three batches are melted together to give an alloy that is 50% copper. Find the weight of Batch C.

Let Batch C weigh y kg. Copper: A = 80 kg, B = 100 kg, C = 0.3y (1½)
0.50 = (80 + 100 + 0.3y)/(300 + y) (1½)
150 + 0.5y = 180 + 0.3y (1) ⇒ 0.2y = 30 ⇒ **y = 150 kg** (1)

35. The perimeter of a rectangular field is 50 m and its area is 150 m². Form a quadratic expression in the breadth x, factorise it and find the length and breadth.

Length + breadth = 25 ⇒ length = 25 − x. Area: $x(25 - x) = 150 \Rightarrow x^2 - 25x + 150 = 0$ (1½)
(x − 10)(x − 15) = 0 (1½)
x = 10 or 15 (1) ⇒ the field is **15 m by 10 m** (1)

OR

The area of a rectangular garden is $(4x^2 + 12x + 5)$ m². Factorise the expression to find its length and breadth. If x = 2, find the perimeter.

$4x^2 + 12x + 5 = 4x^2 + 2x + 10x + 5$ (1½) $= 2x(2x + 1) + 5(2x + 1) = (2x + 1)(2x + 5)$ (1)
Dimensions: **(2x + 1) m and (2x + 5) m** (1)
For x = 2: 5 m and 9 m ⇒ perimeter = 2(5 + 9) = **28 m** (1½)

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. A water tank holds 500 litres. It is emptied at the rate of 20 litres per minute, so the water left after t minutes is W = 500 − 20t litres.

(i) How much water is left after 8 minutes?

W = 500 − 160 = **340 litres**

(ii) Find the zero of the polynomial W = 500 − 20t and say what it means.

500 − 20t = 0 ⇒ t = **25**; the tank is empty after 25 minutes.

(iii) After how many minutes will 120 litres be left?

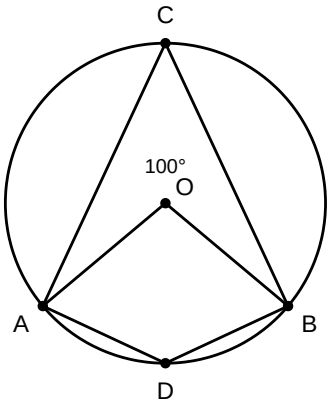
500 − 20t = 120 (1) ⇒ t = **19 minutes** (1)

OR

(iii) Water is poured in at 15 litres per minute while the tank is being emptied at 20 litres per minute. Write the new expression for the water left and find when the tank is empty.

W = 500 − 20t + 15t = 500 − 5t (1) ⇒ empty when t = **100 minutes** (1)

37. On a circular dance floor with centre O, two lights A and B are fixed on the boundary so that the chord AB subtends an angle of 100° at O. A speaker C is placed on the major arc and another speaker D on the minor arc.



(i) Find ∠ACB.

Angle at the circumference = ½ × angle at the centre = **50°**

(ii) Find ∠ADB.

ACBD is a cyclic quadrilateral ⇒ ∠ADB = 180° − 50° = **130°**

(iii) Find ∠OAB.

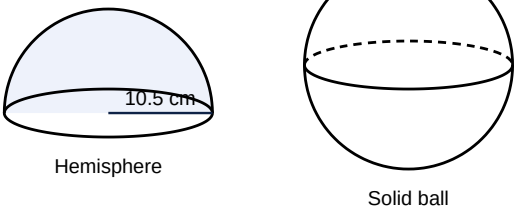
OA = OB ⇒ ∠OAB = ∠OBA (½); ∠OAB = (180° − 100°)/2 (1) = **40°** (½)

OR

(iii) State the relation between ∠ACB and ∠ADB and name the property used.

∠ACB + ∠ADB = 180° (1), because opposite angles of a cyclic quadrilateral are supplementary (1).

38. A solid iron ball has radius 10.5 cm (π = 22/7). It is cut along a plane through its centre into two equal hemispheres.



Hemisphere

Solid ball

(i) Find the surface area of the ball.

$4\pi r^2 = 4 \times 22/7 \times 110.25 =$ **1386 cm²**

(ii) Find the volume of the ball.

$V = 4/3\ \pi r^3 = 4/3 \times 22/7 \times 1157.625 =$ **4851 cm³**

(iii) Find the volume of each hemisphere.

Each hemisphere has half the volume (1): 4851 ÷ 2 = **2425.5 cm³** (1)

O R

(iii) Find the total surface area of each hemisphere.

TSA = CSA + base = $2\pi r^2 + \pi r^2 = 3\pi r^2$ (1) = 3 × 22/7 × 110.25 = **1039.5 cm²** (1)